

Current Literature

Lancet child development series

Series, Child development in developing countries.

Developmental potential in the first 5 years for children in developing countries.

Grantham-McGregor S, Cheung YB, Cueto S, Glewwe P, Richter L, Strupp B, and the International Child Development Steering Group. (2007) *The Lancet*, **369**, 60–70.

Many children younger than 5 years in developing countries are exposed to multiple risks, including poverty, malnutrition, poor health and unstimulating home environments, which detrimentally affect their cognitive, motor and social-emotional development. There are few national statistics on the development of young children in developing countries. We therefore identified two factors with available worldwide data – the prevalence of early childhood stunting and the number of people living in absolute poverty – to use as indicators of poor development. We show that both indicators are closely associated with poor cognitive and educational performance in children, and use them to estimate that over 200 million children under 5 years are not fulfilling their developmental potential. Most of these children live in south Asia and sub-Saharan Africa. These disadvantaged children are likely to do poorly in school and subsequently have low incomes, high fertility and provide poor care for their children, thus contributing to the intergenerational transmission of poverty.

Series, Child development in developing countries. **Child development: risk factors for adverse outcomes in developing countries.**

Walker SP, Wachs TD, Meeks Gardner J, Lozoff B, Wasserman GA, Pollitt E, Carter JA, and the International Child Development Steering Group. (2007) *The Lancet*, **369**, 145–157.

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Poverty and associated health, nutrition and social factors prevent at least 200 million children in developing countries from attaining their developmental potential. We review the evidence linking compromised development with modifiable

biological and psychosocial risks encountered by children from birth to 5 years of age. We identify four key risk factors where the need for intervention is urgent: stunting, inadequate cognitive stimulation, iodine deficiency and iron deficiency anaemia. The evidence is also sufficient to warrant interventions for malaria, intrauterine growth restriction, maternal depression, exposure to violence and exposure to heavy metals. We discuss the research needed to clarify the effect of other potential risk factors on child development. The prevalence of the risk factors and their effect on development and human potential are substantial. Furthermore, risks often occur together or cumulatively, with concomitant increased adverse effects on the development of the world's poorest children.

Series, Child development in developing countries. **Strategies to avoid the loss of developmental potential in more than 200 million children in the developing world.**

Engle PL, Black MM, Behrman JB, Cabral de Mello M, Gertler PJ, Kapiriri L, Martorell R, Eming Young M, and the International Child Development Steering Group. (2007) *The Lancet*, **369**, 229–242.

doi: 10.1016/S0140-6736(07)60112-3

This paper is the third in the Child Development Series. The first paper showed that more than 200 million children under 5 years of age in developing countries do not reach their developmental potential. The second paper identified four well-documented risks: stunting, iodine deficiency, iron deficiency anaemia and inadequate cognitive stimulation, plus four potential risks based on epidemiological evidence: maternal depression, violence exposure, environmental contamination and malaria. This paper assesses strategies to promote child development and to prevent or ameliorate the loss of developmental potential. The most effective early child development programmes provide direct learning experiences to children and families, are targeted towards younger and disadvantaged children, are of longer duration, high quality and high intensity, and are integrated with family support, health, nutrition, or educational systems and services. Despite convincing evidence, programme coverage is low. To achieve the Millennium Development Goals of reducing poverty and ensuring primary school completion for both girls and boys, governments and

civil society should consider expanding high-quality, cost-effective early child development programmes.

These papers comprise another of the Lancet's series on international child health loosely based around the Millennium Development Goals. The first series focused on avoidable mortality in children aged under 5 years (reviewed in Child 2003, issue 29(6)), this series deals with the much greater inequity, at least in terms of numbers, of children failing to meet their developmental potential, and the consequences of this. These include a continuation of the cycle of poverty, losses to the human and economic resources for the country, and increased levels of ill health, disability, malnutrition and other major public health problems. The first paper describes how they calculated that 200 million children fail to meet their potential worldwide. One of the problems here is the lack of internationally comparable data – in fact the lack of data full stop! They have therefore had to make some major assumptions and based their figures on the number of children living in extreme poverty (on less than a dollar a day) and the number of children with stunted growth as a result of malnutrition. One consequence of these assumptions is that it almost certainly underestimates the total number of affected children. They also describe data which suggest that for these 200 million children, the average economic loss is 20% of their potential earnings.

One interesting feature of this paper is their proposed model of impaired child development, which includes three elements of development: sensori-motor, social/emotional, and cognitive/language. This seems a much fuller and simpler description of development than those frequently used by paediatricians and psychologists which often leave out aspects of play, problem solving and social interaction, or tack these on almost as an afterthought.

The second paper discusses risk factors for impaired developmental potential. The striking feature here is the overlap with risk factors for avoidable mortality (i.e. undernutrition and malaria) and with the global burden of disease studies (i.e. malaria again, maternal depression and violence). It is interesting that iodine and iron deficiency are seen as major and important causes of preventable developmental impairment, but this is because of the very large number of children affected. Iodine deficiency affects 35% of people worldwide, and on average, a child brought up with iodine deficiency has an IQ/DQ score of 10–15 points less than a child brought up iodine replete. Iodine supplementation in pregnancy is a key intervention.

In general, intervention studies have a moderate effect of roughly 0.5 standard deviations – some a bit more or some a bit less – but these are huge in public health terms. In addition, most children suffer combined risk factors which have a cumulative effect; hence addressing multiple risk factors would have a commensurately large effect and would be a more efficient approach, especially as many of these are nutritional.

The third paper describes the characteristics of effective interventions. The best evidence is for early combined child development programmes. There is an interesting comparison here with the rather lukewarm results of the early evaluations of the

UK Sure Start programme. However, the emphasis on nutrition, both as a risk factor and as a focus for intervention, is much more important in the international studies. There is less direct evidence for the benefits of reducing social, environmental and infection risks, but in view of the other beneficial outcomes, these programmes are supported. The paper concludes by arguing for greater investment in child development programmes for human rights reasons, economic benefits, and to reduce disparities between countries.

Overall, this series is an important source of reference and material for advocacy. Early interventions promote child development and prevent developmental impairment. Despite this, the response from centres of power has been slow. Children's rights are violated, states' and individuals' economic prosperity is hampered, and the Millennium Development Goals are threatened. Improving child development is the other side of the same coin of reducing avoidable mortality.

Richard Reading

Assessing the burden of injury in six European countries.

Polinder S, Meerdink WJ, Mulder S, Petridou E, van Beeck E, and EUROCOST Reference Group. (2007) *Bulletin of the World Health Organization*, 85, 27–34.

Objective To assess injury-related mortality, disability and disability-adjusted life years (DALYs) in six European countries. **Methods** Epidemiological data (hospital discharge registers, emergency department registers, mortality databases) were obtained for Austria, Denmark, Ireland, Netherlands, Norway and the United Kingdom (England and Wales). For each country, the burden of injury was estimated in years lost due to premature mortality (YLL), years lived with disability (YLD), and DALYs (per 1000 persons).

Findings We observed marked differences in the burden of injury between countries. Austria lost the largest number of DALYs (25 per 1000 persons), followed by Denmark, Norway and Ireland (17–20 per 1000 persons). In the Netherlands and the United Kingdom, the total burden due to injuries was relatively low (12 per 1000 persons). The variation between countries was attributable to a high variation in premature mortality (YLL varied from 9 to 17 per 1000 persons) and disability (YLD varied from 2 to 8 per 1000 persons). In all countries, males aged 25–44 years represented one-third of the total injury burden, mainly due to traffic and intentional injuries. Spinal cord injury and skull-brain injury resulted in the highest burden due to permanent disability.

Conclusion The burden of injury varies considerably among the six participating European countries, but males aged 15–24 years are responsible for a disproportionate share of the

assessed burden of injury in all countries. Consistent injury control policy is supported by high-quality summary measures of population health. There is an urgent need for standardized data on the incidence and functional consequences of injury.

This study provides very useful information on the burden of injuries in terms of disability and mortality. Remarkably, hitherto this information has been either unavailable or patchy. Inevitably there are weaknesses and deficiencies in some of the data, but these are well described in the paper and tend to underestimate the overall burden rather than overestimate it. The data are for all ages but are disaggregated by age group, including children. The one major implication of the study is for better comparable data; nevertheless the burdens are particularly notable in adolescents and young adults, especially males. The study highlights the relative lack of research investment in a subject with huge public health and economic consequences.

Richard Reading

Operational implications of using 2006 World Health Organization growth standards in nutrition programmes: secondary data analysis.

Seal A, Kerac M. (2007) *British Medical Journal*, 334, 733–735. doi: 10.1136/bmj.39101.664109.AE

Objective To assess the implications of adopting the World Health Organization (WHO) 2006 growth standards in combination with current diagnostic criteria in emergency and non-emergency child feeding programmes.

Design Secondary analysis of data from three standardized nutrition surveys ($n = 2555$) for prevalence of acute malnutrition, using weight-for-height z-score (<-2 and <-3) and percentage of the median ($<80\%$ and $<70\%$) cut-offs for moderate and severe acute malnutrition from the National Center for Health Statistics/WHO growth reference (NCHS reference) and the new WHO 2006 growth standards (WHO standards).

Setting Refugee camps in Algeria, Kenya and Bangladesh.

Population Children aged 6–59 months.

Results Important differences exist in the weight-for-height cut-offs used for defining acute malnutrition obtained from the WHO standards and NCHS reference data. These vary according to a child's height and according to whether z-score or percentage of the median cut-offs are used. If applied and used according to current practice in nutrition programmes, the WHO standards will result in a higher measured prevalence of severe acute malnutrition during surveys but, paradoxically, a decrease in the admission of children to emergency feeding programmes and earlier discharge of recovering patients. The

expected impact on case fatality rates of applying the new standards in conjunction with current diagnostic criteria is unknown.

Conclusions A full assessment of the appropriate use of the new WHO standards in the diagnosis of acute malnutrition is urgently needed. This should be completed before the standards are adopted by organizations that run nutrition programmes targeting acute malnutrition.

There is currently debate and discussion about whether or not to introduce the WHO growth standards into UK practice. This paper considers the question from an international perspective, and particularly looks at the implications for countries and children within them with high levels of malnutrition. The point the authors make is similar to those made on the introduction of any new standards (for example, similar points were made when the current UK standards were introduced in the early 1990s), but in this case, the implications are potentially much more significant. The authors suggest that much higher rates of malnutrition may be reported from countries, but paradoxically far fewer children will meet the criteria for being included in feeding programmes. The obvious answer is to redefine both sets of criteria, but in e-mail discussions of this paper, the authors point out that what might seem simple and sensible to academics and policymakers, may be much less clear to front-line staff on the ground, who have been used to the current simple definitions for a long time and may find it hard to change.

If you are interested in this matter, then this is an important paper, not necessarily as a definitive statement, but as raising an important issue and stimulating debate. Unfortunately, it is now too late to enter into the UK government consultation on this question, but it is important to know about international implications of a policy which may be implemented locally.

Richard Reading

A continuous-scale measure of child development for population-based epidemiological surveys: a preliminary study using Item Response Theory for the Denver Test.

Drachler ML, Marshall T, de Carvalho Leite JC. (2007) *Paediatric and Perinatal Epidemiology*, 21, 138–153.

A method for translating research data from the Denver Test into individual scores of developmental status measured in a continuous scale is presented. It was devised using the Denver Developmental Screening Test (DDST) but can be used for Denver II. The DDST was applied in a community-based survey of 3389 under-5-year-olds in Porto Alegre, Brazil. The items of success were standardized by logistic regression on log chronological age. Each child's ability age was then estimated by maximum likelihood as the age in this reference population

corresponding to the child's success and failures in the test. The score of developmental status is the natural logarithm of this ability age divided by chronological age, and thus measures the delay or advance in the child's ability age compared with chronological age. This method estimates development status using both difficulty and discriminating power of each item in the reference population, an advantage over scores based on total number of items correctly performed or failed, which depend on difficulty only. The score corresponds with maternal opinion of child developmental status and with the three-category scale of the DDST. It shows good construct validity, indicated by symmetrical and homogeneous variability from 3 months upwards, and reasonable results in describing gender differences in development by age, the mean score increasing with socio-economic conditions and diminishing among low-birthweight children. If a standardized measure of development status (z-scores) is required, this can be obtained by dividing the score by its standard deviation. Concurrent and discriminant validity of the score must be examined in further studies.

This is a very detailed and preliminary study into an extension of the Denver Developmental Screening Test for use as an epidemiological tool. As such, it should be of interest. The main criticisms of the Denver test as it stands are that it is essentially based on negatives. The outcomes are passing and failing, broadly speaking although there are two other categories of borderline and untestable. All of these depend on how many test items are completed. There is no means of assessing developmental age, nor even of identifying children who are developing more rapidly than the developmental norms. Furthermore, there has been criticism of the appropriateness of these developmental norms in different settings.

This is disappointing, as the test can be completed by self-report, or rapidly by questioning and observation by an otherwise unqualified assessor – both important properties of an epidemiological tool especially in resource-poor countries.

This study describes a method of incorporating the different test items into a continuous scale. Only time and further validation studies will tell whether this is a useful approach, but if it does work, it would be extremely valuable in a wide range of research applications.

Richard Reading

Role of home visiting in improving parenting and health in families at risk of abuse and neglect: results of a multicentre randomised controlled trial and economic evaluation.

Barlow J, Davis H, McIntosh E, Jarrett P, Mockford C, Stewart-Brown S. (2007) *Archives of Disease in Childhood*, 92, 229–233.

doi: 10.1136/adc.2006.095117

Objectives To evaluate the effectiveness and cost-effectiveness of an intensive home visiting programme in improving outcomes for vulnerable families.

Design Multicentre randomized controlled trial in which eligible women were allocated to receive home visiting ($n = 67$) or standard services ($n = 64$). Incremental cost analysis.

Setting 40 general practitioner practices across two counties in the UK.

Participants 131 vulnerable pregnant women.

Intervention Selected health visitors were trained in the Family Partnership Model to provide a weekly home visiting service from 6 months antenatally to 12 months post-natally.

Main outcome measures Mother–child interaction, maternal psychological health attitudes and behaviour, infant functioning and development, and risk of neglect or abuse.

Results At 12 months, differences favouring the home-visited group were observed on an independent assessment of maternal sensitivity ($P < 0.04$) and infant cooperativeness ($P < 0.02$). No differences were identified on any other measures. A non-significant increase in the likelihood of intervention group infants being the subject of child protection proceedings, or being removed from the home, and one death in the control group were found. The mean incremental cost per infant of the home visiting intervention was £3246 (bootstrapped 95% CI for the difference £1645–4803).

Conclusion This intervention may have the potential to improve parenting and increase the identification of infants at risk of abuse and neglect in vulnerable families. Further investigation is needed, along with long-term follow-up to assess possible sleeper effects.

This was a very intensive home visiting programme of weekly visits for 18 months, albeit in a high-risk group of families. The evaluation was well conducted, with the main weakness being a possible lack of statistical power due to insufficient numbers of participants. Although the abstract talks up the results, the overall impression is of modest effects only, which may have occurred by chance. In the paper, the authors acknowledge that this is insufficient evidence to support the introduction of such an intensive intervention on a widespread scale. Nevertheless, the opportunity for longer-term follow-up of these two comparison groups is possible (see comment to the following two papers), and it will be interesting to see if any differences between the intervention and control groups start to become evident in later childhood, perhaps as a result of small changes in attachment which might amplify over childhood.

Richard Reading

Parenting intervention in Sure Start services for children at risk of developing conduct disorder: pragmatic randomised controlled trial.

Hutchings J, Bywater T, Daley D, Gardner F, Whitaker C, Jones K, Eames C, Edwards RT. (2007) *British Medical Journal*, 334, 678–682.
doi: 10.1136/bmj.39126.620799.55

Objective To evaluate the effectiveness of a parenting programme as a preventive intervention with parents of preschool children considered to be at risk of developing conduct disorder.

Design Pragmatic randomized controlled trial using a block design with allocation by area.

Setting Eleven Sure Start areas in north and mid-Wales.

Participants 153 parents from socially disadvantaged areas, with children aged 36–59 months at risk of conduct disorder defined by scoring over the clinical cut-off on the Eyberg child behaviour inventory. Participants were randomized on a 2:1 basis, 104 to intervention and 49 to remaining on the wait listing (control). Twenty (13%) were lost to follow-up 6 months later, 18 from the intervention group.

Intervention The Webster-Stratton Incredible Years basic parenting programme, a 12-week group-based intervention.

Main outcome measures Problem behaviour in children and parenting skills assessed by self-reports from parents and by direct observation in the home. Parents' self-reported parenting competence, stress and depression. Standardized and well-validated instruments were used throughout.

Results At follow-up, most of the measures of parenting and problem behaviour in children showed significant improvement in the intervention group. The intention-to-treat analysis for the primary outcome measure, the Eyberg child behaviour inventory, showed a mean difference between groups of 4.4 points (95% confidence interval 2.0–6.9, $P < 0.001$) on the problem scale with an effect size of 0.63, and a mean difference of 25.1 (14.9–35.2, $P < 0.001$) on the intensity scale with an effect size of 0.89.

Conclusion This community-based study showed the effectiveness of an evidence-based parenting intervention delivered with fidelity by regular Sure Start staff. It has influenced policy within Wales and provides lessons for England, where, to date, Sure Start programmes have not been effective.

Parenting programme for parents of children at risk of developing conduct disorder: cost effectiveness analysis.

Edwards RT, C  illeachair A, Bywater T, Hughes DA, Hutchings J. (2007) *British Medical Journal*, 334, 682–685.
doi: 10.1136/bmj.39126.699421.55

Objective To investigate the cost-effectiveness of a parenting programme.

Design An incremental cost-effectiveness analysis alongside a pragmatic randomized controlled trial of the effectiveness of a group-parenting programme delivered through Sure Start in the community.

Setting Sure Start areas in north and mid-Wales.

Participants Parents of 116 children aged 36–59 months (87% of the clinical sample) at risk of developing conduct disorders (defined by scoring over the clinical cut-off on the Eyberg child behaviour inventory). Children were identified by health visitors and recruited by the research team.

Intervention The Webster-Stratton Incredible Years basic parenting programme or a 6-month waiting list control.

Main outcome measure Incremental cost per unit of improvement on the intensity score of the Eyberg child behaviour inventory.

Results The bootstrapped incremental cost-effectiveness ratio point estimate was £73 (  109, \$142) per one point improvement on the intensity score (95% confidence interval   42 to   140). It would cost   5486 (  8190, \$10 666) to bring the child with the highest intensity score to below the clinical cut-off point and   1344 (  2006, \$2618) to bring the average child in the intervention group within the non-clinical limits on the intensity score (below 127). For a ceiling ratio of   100 (  149, \$194) per point increase in intensity score, there is an 83.9% chance of the intervention being cost-effective. The mean cost per child attending the parenting group was   1934 (  2887, \$3760) for eight children and   1289 (  1924, \$2506) for 12 children, including initial costs and materials for training group leaders. When we categorized the sample into relatively mild, moderate and severe behaviour groups based on intensity scores at baseline, the intervention seemed more cost-effective in those with the highest risk of developing conduct disorder.

Conclusion This parenting programme improves child behaviour as measured by the intensity score of the Eyberg child behaviour inventory at a relatively low cost and was cost-effective compared with the waiting list control. This parenting programme involves modest costs and demonstrates strong clinical effect, suggesting it would represent good value for money for public spending.

Meanwhile, in the BMJ, there is a positive trial of a behavioural intervention to reduce risk factors for later conduct disorder in young children. The Incredible Years programme is more focused on enhancing appropriate behaviour management than on relationship building, and has been shown to be effective in carefully controlled clinical trials. The importance of this study is that it was conducted in a pragmatic way and examined the

effect of the intervention when applied in a standard clinical setting. This contrasts with a number of other parenting interventions shown effective in controlled settings which are unsuccessful when transferred into the real world. A good example of the latter is the whole Sure Start programme in England, which was based on the best evidence available, yet which a formal evaluation found to have only modest benefit and even to have paradoxical disbenefits among the most vulnerable children (see Belsky J, Melhuish E, Barnes J, Leyland AH, Romaniuk H. (2006) Effects of Sure Start local programmes on children and families: early findings from a quasi-experimental, cross sectional study. *British Medical Journal*, 332, 1476–1478).

This trial, which appears to have robust methods, shows a clear benefit over a range of prognostic risk factors, such as emotional problems, hyperactive behaviour, and self control in the child; stress, depression and parenting behaviour in the mother, and emotional problems in the siblings. Some of these measures were based on parental questionnaire, and so subject to possible bias, but others were more objective, based on home observations by researchers blinded to the trial allocation. The beneficial effects on siblings suggest the improved parenting was generalized to other children in the family.

The main weakness of the trial is that outcomes were measured after a mere 3 months from the end of the parenting course. Longer-term comparisons have been ruled out because after this time, all the control families were offered the parenting course.

The accompanying paper on the cost-effectiveness of the intervention shows relatively modest costs for substantial benefits in the short term. In other words, this looks like good value for investment of public money. However, this analysis depends on the benefits of the intervention being sustained, and on the improvement in risk factors being translated into reductions in subsequent conduct disorder. If this occurs, then the cost-benefit relationship could be very great, but this study, in common with many other randomized trials of parenting interventions, does not allow estimation of these longer-term outcomes or costs, because control families are offered the intervention.

Why is this methodology followed so often? It does not happen in drug trials, and it betrays an underlying assumption of the researchers that parenting interventions are always potentially beneficial, and never have adverse effects. We need comparisons between intervention and control groups over a much longer time in these trials in order to be able to estimate the real effect of the interventions. Some may be 'slow burners' and have small initial benefits which are not shown in the early comparisons, but which operate cumulatively. Others may be 'Roman candles', which go off with a bang, burn brightly for a while but then fade away to nothing. We have no way of telling with these studies; the irony is that the impact on children, families and society may be far greater than any drug or conventional medical technology.

Richard Reading